Beyond Gradual Warming

European Food-and-Energy Exposure under CMIP6 Future Projections and Severe AMOC Weakening Stress Tests

Thesis proposal presentation

May 2026

Beyond Gradual Warming — Food-and-Energy Exposure under AMOC Stress Tests

Motivation. Portmann et al. (2026): $51 \pm 8\%$ AMOC weakening by 2100 under SSP2-4.5 vs. multi-model mean. Xing et al. (2026): two-decade decline in the observed western-boundary current.

RQ. To what extent would *severe AMOC weakening* generate European agricultural and household-energy exposure *beyond* the risks implied by CMIP6 future climate projections?

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Inputs & estimation

Crop panel | Energy-demand panel | historical crop–weather indicators & seasonal weather regressors.

Crop model

$$\begin{aligned}\ln(Yield_{c,k,t}) = \beta_{1k} GDD_{c,k,t} + \beta_{2k} Heat_{c,k,t} + \beta_{3k} Frost_{c,k,t} \\ + \beta_{4k} Precip_{c,k,t} + \beta_{5k} Precip^2_{c,k,t} + \alpha_{c,k} + \lambda_t + \epsilon_{c,k,t}\end{aligned}$$

Energy model

$$\begin{aligned}\ln(Energy_{c,f,t}) = \beta_H HDD^{heat}_{c,t} + \beta_C CDD^{cool}_{c,t} \\ + \beta_G \ln(Price_{c,f,t}) + \alpha_c + \lambda_t + \epsilon_{c,f,t}\end{aligned}$$

Estimate response functions, then compare scenarios *with* (NAHosMIP AMOC anomaly[Multiple models]) vs. *without*.

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Outputs & expected results

For each EU country & Switzerland, at NUTS2 level:

- Crop exposure
- Household Energy exposure
- Total exposure

Mapped as the *incremental* risk attributable to severe AMOC weakening beyond CMIP6 projections.

Limitations: Scenario – stress test, not forecast since: Idealised experiment and no adaptation; Responses – Not able to capture if going outside of the historical range; Resolution – NUTS2 may mask local heterogeneity.

Appendix

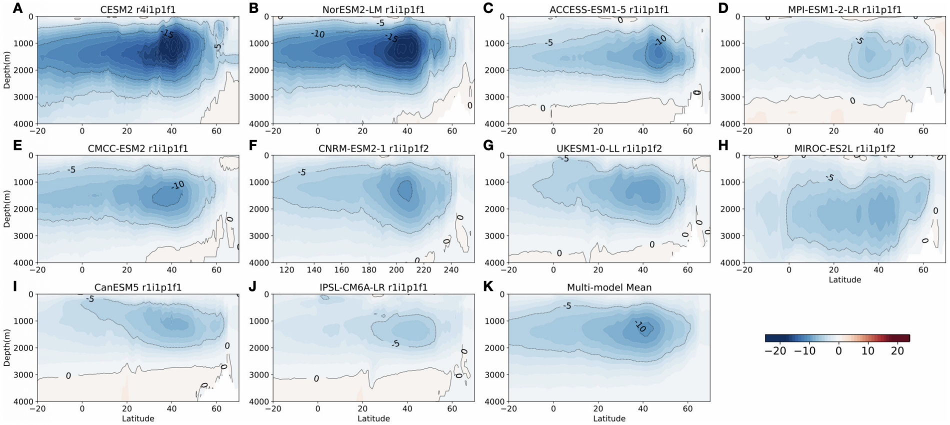

FIGURE 3

Maps of AMOC difference (in *Sv*) between 31-year averages over 2070-2100 in the SSP585 scenario and 1970-2000 in the historical runs for ten models **(A-J)** and the the multi-model mean **(K)**, visualized as basin-wide zonal-and time-integrated stream function.

Figure: Zhang,Ito, Bracco 2024

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